3.2.S.7.1 Stability Summary and Conclusions

The retest data at the recommended storage condition of -20 ± 5°C for 2439-exNA drug substance batch XXX meets the specifications listed in Table 2. As can be seen in the data presented in Section 3.2.S.7.3, no significant changes are noted in the results from any test method. This confirms that when drug product batch # XXX was manufactured the drug substance batch XXX met specification.

3.2.S.7.2 Introduction

A single batch of GMP drug substance, XXX, manufactured at Shanghai Hongene Bioengineering Co., Ltd. had aliquots stored at the long-term storage condition of -20 ± 5°C. The aliquots are retested at key points to support drug product manufacture., such as following manufacture of drug product. This approach brackets Details of the retest parameters for batch XXX are provided in Table 1.

Table 1 2439-exNA Drug Substance Subject to Retest

Lot Number	Manufacturing Date	Container Closures	Storage Conditions
XXX	March 2024	Primary packaging: Nalgene Bottles, PETG, 60 mL with 24 mm HDPE caps. Secondary Packaging: Aluminum foil bag, sealed by heat. Tertiary packaging: Desiccant pouch in aluminum foil bag, sealed by heat.	–20 ± 5°C

PETG – PolyEthylene Terephthalate Glycol; HDPE – High Density PolyEthylene

3.2.S.7.3 Specification for retest

Retesting is performed as per the specification provided in Table 2

Table 2 2439-exNA Drug Substance Stability Specification

Test	Method	Acceptance Criteria
Appearance	Visual	Colorless to pale yellow solid
Assay	UV-Visible Spectrophotometry	Report Value
Purity and impurities	Denaturing IEX-HPLC	Sense strand: Report value Antisense strand: Report value Report ≥0.2% single impurity
Purity and impurities	Non-Denaturing SEC-HPLC	Report Report ≥0.2% single impurity
Water Content	Karl Fischer, USP <921>	≤15 % w/w
Bacterial	USP <85>, Gel clot method	NMT 0.05 EU/mg

Table 2 2439-exNA Drug Substance Stability Specification

Test	Method	Acceptance Criteria
Endotoxin		

EU – endotoxin unit; SEC – Size Exclusion Chromatography; IEX – Ion Exchange chromatography; UV – Ultraviolet

3.2.S.7.4 Retesting plan

Two aliquots of 2439-exNA drug substance from batch # XXX were retained at the long-term storage condition of -20 ± 5°C to retest after the drug product manufacture. One aliquot was retested after the fill date of the 24th June 2024 for drug product batch # XXX where batch XXX was formulated and sterile fill/finished for use in the Phase 1/2 study. One aliquot continues to be retained for future testing.

3.2.S.7.5 Retest Period

At this point in the development lifecycle, no retest period is claimed. 2439-exNA drug substance batch XXX was consumed in the manufacture of GMP drug product batch # XXX. Drug substance batch XXX met specifications at release and after 5 months storage at -20 ± 5°C which brackets the date of formulation of
XXX into drug product.